

JESS Thermodynamic Database v8.9

Chemical species in reactions with Curium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
2OH2MePropan-1				
-1	---	197	C(4)H(7)O(3)	103.098
2ThenTriFAcone-1				
-1	---	101	C(8)H(4)F(3)O(2)S(1)	221.174
Acetic-1 Acetate ion				
-1	71-50-1	558	C(2)H(3)O(2)	59.0446
Br-1 Bromide ion				
-1	24959-67-9	1213	Br(1)	79.9040
C(s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C(1)	12.0110
Ca(s) Calcium; Calcium, cubic; Calcium, fcc; Calcium, alpha				
0	7440-70-2	184	Ca(1)	40.0780
CDTA-4 trans-1,2-cyclohexylenedinitrilotetraacetate ion; trans-1,2-diaminocyclohexanetetraacetate ion; CDTA ion				
-4	22005-54-5	301	C(14)H(18)N(2)O(8)	342.306
Citric-3 Citrate ion				
-3	126-44-3	1347	C(6)H(5)O(7)	189.102
Cl-1 Chloride ion				

-1	16887-00-6	2242	Cl (1)	35.4530
Cl2 (g) Chlorine gas				
0	7782-50-5	589	Cl (2)	70.9060
Cm+2				
2	---	2	Cm (1)	247.000
Cm+3 Curium(III) ion				
3	22541-42-0	162	Cm (1)	247.000
Cm+3_2OH2MePropan-1				
2	---	2	C (4) H (7) Cm (1) O (3)	350.098
Cm+3_2OH2MePropan-1 (2)				
1	---	3	C (8) H (14) Cm (1) O (6)	453.196
Cm+3_2OH2MePropan-1 (3)				
0	---	2	C (12) H (21) Cm (1) O (9)	556.293
Cm+3_2ThenTriFAcone-1 (3)				
0	---	1	C (24) H (12) Cm (1) F (9) O (6) S (3)	910.521
Cm+3_Acetic-1				
2	---	2	C (2) H (3) Cm (1) O (2)	306.045
Cm+3_Acetic-1 (2)				
1	---	2	C (4) H (6) Cm (1) O (4)	365.089
Cm+3_Acetic-1 (3)				
0	---	1	C (6) H (9) Cm (1) O (6)	424.134
Cm+3_Acetic-1 (4)				
-1	---	1	C (8) H (12) Cm (1) O (8)	483.179
Cm+3_Br-1				

2	---	1	Br (1) Cm (1)	326.904
Cm+3_Ca+2_OH-1 (3)				
2	---	1	Ca (1) Cm (1) H (3) O (3)	338.100
Cm+3_Ca+2 (2)_OH-1 (4)				
3	---	1	Ca (2) Cm (1) H (4) O (4)	395.185
Cm+3_Ca+2 (3)_OH-1 (6)				
3	---	1	Ca (3) Cm (1) H (6) O (6)	469.278
Cm+3_CDTA-4				
-1	---	1	C (14) H (18) Cm (1) N (2) O (8)	589.306
Cm+3_Citric-3				
0	---	3	C (6) H (5) Cm (1) O (7)	436.102
Cm+3_Citric-3 (2)				
-3	---	3	C (12) H (10) Cm (1) O (14)	625.203
Cm+3_Cl-1				
2	---	3	Cl (1) Cm (1)	282.453
Cm+3_Cl-1_OH-1 (3)				
-1	---	1	Cl (1) Cm (1) H (3) O (3)	333.475
Cm+3_Cl-1 (2)				
1	---	2	Cl (2) Cm (1)	317.906
Cm+3_CO3-2				
1	---	3	C (1) Cm (1) O (3)	307.009
Cm+3_CO3-2 (2)				
-1	---	4	C (2) Cm (1) O (6)	367.018
Cm+3_CO3-2 (3)				
-3	---	4	C (3) Cm (1) O (9)	427.028

Cm+3_CO3-2 (4)				
-5	---	3	C (4) Cm (1) O (12)	487.037
Cm+3_CO3-2 (5)				
-7	---	1	C (5) Cm (1) O (15)	547.046
Cm+3_DTPA-5				
-2	---	1	C (14) H (18) Cm (1) N (3) O (10)	635.311
Cm+3_EDTA-4				
-1	---	2	C (10) H (12) Cm (1) N (2) O (8)	535.214
Cm+3_EDTMP-8				
-5	---	1	C (6) H (12) Cm (1) N (2) O (12) P (4)	675.064
Cm+3_Etbis (ImMePhos) -4				
-1	---	1	C (4) H (10) Cm (1) N (2) O (6) P (2)	491.081
Cm+3_F-1				
2	---	3	Cm (1) F (1)	265.998
Cm+3_F-1 (2)				
1	---	4	Cm (1) F (2)	284.997
Cm+3_F-1 (3)				
0	---	3	Cm (1) F (3)	303.995
Cm+3_F*3Benzoylacetone-1 (3)				
0	---	1	C (30) H (18) Cm (1) F (9) O (6)	892.455
Cm+3_Glycolic-1				
2	---	2	C (2) H (3) Cm (1) O (3)	322.044
Cm+3_Glycolic-1 (2)				
1	---	2	C (4) H (6) Cm (1) O (6)	397.088

Cm+3_H+1_CDTA-4				
0	---	1	C (14) H (19) Cm (1) N (2) O (8)	590.314
Cm+3_H+1_Citric-3				
1	---	3	C (6) H (6) Cm (1) O (7)	437.109
Cm+3_H+1_Citric-3 (2)				
-2	---	4	C (12) H (11) Cm (1) O (14)	626.211
Cm+3_H+1_CO3-2				
2	---	1	C (1) H (1) Cm (1) O (3)	308.017
Cm+3_H+1_DTPA-5				
-1	---	1	C (14) H (19) Cm (1) N (3) O (10)	636.319
Cm+3_H+1_EDTA-4				
0	---	3	C (10) H (13) Cm (1) N (2) O (8)	536.222
Cm+3_H+1_EDTMP-8				
-4	---	1	C (6) H (13) Cm (1) N (2) O (12) P (4)	676.071
Cm+3_H+1_Etbis (ImMePhos) -4				
0	---	1	C (4) H (11) Cm (1) N (2) O (6) P (2)	492.089
Cm+3_H+1_Gly-1				
3	---	2	C (2) H (5) Cm (1) N (1) O (2)	322.067
Cm+3_H+1_HIMDA-2				
2	---	1	C (6) H (10) Cm (1) N (1) O (5)	423.149
Cm+3_H+1_HOEDTA-3 (2)				
-2	---	1	C (20) H (31) Cm (1) N (4) O (14)	798.485
Cm+3_H+1_HydrazineNNDiAcet-2				
2	---	1	C (4) H (7) Cm (1) N (2) O (4)	394.111

Cm+3_H+1_MePhos-2				
2	---	1	C (1) H (4) Cm (1) O (3) P (1)	342.015
Cm+3_H+1_N2COOEtIDA-3				
1	---	2	C (7) H (9) Cm (1) N (1) O (6)	450.152
Cm+3_H+1_NTA-3 (2)				
-2	---	1	C (12) H (13) Cm (1) N (2) O (12)	624.242
Cm+3_H+1_OHMePhos-2				
2	---	1	C (1) H (4) Cm (1) O (4) P (1)	358.014
Cm+3_H+1_PhosAcet-3				
1	---	1	C (2) H (3) Cm (1) O (5) P (1)	385.017
Cm+3_H+1 (2)_EDDPI-2				
3	---	1	C (4) H (14) Cm (1) N (2) O (4) P (2)	463.114
Cm+3_H+1 (2)_EDTMP-8				
-3	---	1	C (6) H (14) Cm (1) N (2) O (12) P (4)	677.079
Cm+3_H+1 (2)_Etbis (ImMePhos) -4				
1	---	1	C (4) H (12) Cm (1) N (2) O (6) P (2)	493.097
Cm+3_H+1 (2)_HIMDA-2 (2)				
1	---	1	C (12) H (20) Cm (1) N (2) O (10)	599.298
Cm+3_H+1 (2)_HOEDTA-3 (2)				
-1	---	1	C (20) H (32) Cm (1) N (4) O (14)	799.493
Cm+3_H+1 (2)_OHMePhos-2 (2)				
1	---	1	C (2) H (8) Cm (1) O (8) P (2)	469.029
Cm+3_H+1 (2)_PhosAcet-3				
2	---	1	C (2) H (4) Cm (1) O (5) P (1)	386.025

Cm+3_H+1 (2) _PhosAcet-3 (2)				
-1	---	1	C (4) H (6) Cm (1) O (10) P (2)	523.034
Cm+3_H+1 (2) _PO4-3				
2	---	1	Cm (1) H (2) O (4) P (1)	343.988
Cm+3_H+1 (3) _EDTMP-8				
-2	---	1	C (6) H (15) Cm (1) N (2) O (12) P (4)	678.087
Cm+3_H+1 (3) _Etbis (ImMePhos) -4				
2	---	1	C (4) H (13) Cm (1) N (2) O (6) P (2)	494.105
Cm+3_H+1 (4) _EDTMP-8				
-1	---	1	C (6) H (16) Cm (1) N (2) O (12) P (4)	679.095
Cm+3_H+1 (5) _EDTMP-8				
0	---	1	C (6) H (17) Cm (1) N (2) O (12) P (4)	680.103
Cm+3_HIMDA-2				
1	---	1	C (6) H (9) Cm (1) N (1) O (5)	422.141
Cm+3_HIMDA-2 (2)				
-1	---	1	C (12) H (18) Cm (1) N (2) O (10)	597.282
Cm+3_HOEDTA-3				
0	---	1	C (10) H (15) Cm (1) N (2) O (7)	522.238
Cm+3_HOEDTA-3 (2)				
-3	---	1	C (20) H (30) Cm (1) N (4) O (14)	797.477
Cm+3_HydrazineNNDiAcet-2				
1	---	1	C (4) H (6) Cm (1) N (2) O (4)	393.103
Cm+3_HydrazineNNDiAcet-2 (2)				
-1	---	1	C (8) H (12) Cm (1) N (4) O (8)	539.205

Cm+3_Lactic-1				
2	---	2	C (3) H (5) Cm (1) O (3)	336.071
Cm+3_Lactic-1 (2)				
1	---	2	C (6) H (10) Cm (1) O (6)	425.142
Cm+3_Lactic-1 (3)				
0	---	1	C (9) H (15) Cm (1) O (9)	514.213
Cm+3_MeEDTA-4				
-1	---	1	C (11) H (14) Cm (1) N (2) O (8)	549.241
Cm+3_N2COOEtIDA-3				
0	---	2	C (7) H (8) Cm (1) N (1) O (6)	449.144
Cm+3_N2COOEtIDA-3 (2)				
-3	---	1	C (14) H (16) Cm (1) N (2) O (12)	651.287
Cm+3_N2COOPhenIDA-3				
0	---	1	C (11) H (8) Cm (1) N (1) O (6)	497.188
Cm+3_N2PyridMeAsp-2				
1	---	1	C (10) H (10) Cm (1) N (2) O (4)	469.200
Cm+3_N2PyridMeAsp-2 (2)				
-1	---	1	C (20) H (20) Cm (1) N (4) O (8)	691.401
Cm+3_N3-1				
2	---	1	Cm (1) N (3)	289.020
Cm+3_Na+1_CO3-2 (2) _H2O (6) _ (s)				
0	---	1	C (2) H (12) Cm (1) Na (1) O (12)	498.100
Cm+3_NO2-1				
2	---	1	Cm (1) N (1) O (2)	293.006

Cm+3_NO3-1				
2	---	2	Cm(1)N(1)O(3)	309.005
Cm+3_NO3-1(2)				
1	---	1	Cm(1)N(2)O(6)	371.010
Cm+3_NTA-3				
0	---	2	C(6)H(6)Cm(1)N(1)O(6)	435.117
Cm+3_NTA-3(2)				
-3	---	2	C(12)H(12)Cm(1)N(2)O(12)	623.234
Cm+3_OH-1				
2	---	6	Cm(1)H(1)O(1)	264.007
Cm+3_OH-1_Citric-3				
-1	---	1	C(6)H(6)Cm(1)O(8)	453.109
Cm+3_OH-1_Citric-3(2)				
-4	---	1	C(12)H(11)Cm(1)O(15)	642.210
Cm+3_OH-1_CO3-2_(s)				
0	---	1	C(1)H(1)Cm(1)O(4)	324.017
Cm+3_OH-1(2)				
1	---	4	Cm(1)H(2)O(2)	281.015
Cm+3_OH-1(2)_Citric-3				
-2	---	1	C(6)H(7)Cm(1)O(9)	470.116
Cm+3_OH-1(3)				
0	---	2	Cm(1)H(3)O(3)	298.022
Cm+3_OH-1(3)_(am.,s) Curium hydroxide, amorphous				
0	---	4	Cm(1)H(3)O(3)	298.022

Cm+3_OH-1(3)_ (s) Curium hydroxide				
0	---	3	Cm(1)H(3)O(3)	298.022
Cm+3_Oxalic-2				
1	---	2	C(2)Cm(1)O(4)	335.020
Cm+3_Oxalic-2(2)				
-1	---	2	C(4)Cm(1)O(8)	423.039
Cm+3_Oxalic-2(3)				
-3	---	1	C(6)Cm(1)O(12)	511.059
Cm+3_P3O9-3				
0	---	1	Cm(1)O(9)P(3)	483.917
Cm+3_PO4-3				
0	---	2	Cm(1)O(4)P(1)	341.972
Cm+3_PO4-3(2)				
-3	---	1	Cm(1)O(8)P(2)	436.943
Cm+3_PO4-3(3)				
-6	---	1	Cm(1)O(12)P(3)	531.915
Cm+3_PO4-3(4)				
-9	---	1	Cm(1)O(16)P(4)	626.886
Cm+3_SCN-1				
2	---	3	C(1)Cm(1)N(1)S(1)	305.078
Cm+3_SCN-1(2)				
1	---	2	C(2)Cm(1)N(2)S(2)	363.155
Cm+3_SCN-1(3)				
0	---	1	C(3)Cm(1)N(3)S(3)	421.233

Cm+3_SO4-2				
1	---	4	Cm(1)O(4)S(1)	343.058
Cm+3_SO4-2(2)				
-1	---	3	Cm(1)O(8)S(2)	439.115
Cm+3_SO4-2(3)				
-3	---	2	Cm(1)O(12)S(3)	535.173
Cm+3_Squaric-2				
1	---	1	C(4)Cm(1)O(4)	359.042
Cm+3_Squaric-2(2)				
-1	---	1	C(8)Cm(1)O(8)	471.083
Cm+3_Tartaric-2(2)				
-1	---	1	C(8)H(8)Cm(1)O(12)	543.144
Cm+3_TriMDTA-4				
-1	---	1	C(11)H(14)Cm(1)N(2)O(8)	549.241
Cm+3(2)_CO3-2(3)_(s)				
0	---	1	C(3)Cm(2)O(9)	674.028
Cm+3(2)_H+1(2)_Cl-1_CO3-2(2)				
3	---	1	C(2)H(2)Cl(1)Cm(2)O(6)	651.487
Cm+4				
4	---	14	Cm(1)	247.000
Cm(s) Curium				
0	7440-51-9	28	Cm(1)	247.000
Cm2O3(s) Curium oxide				
0	---	1	Cm(2)O(3)	541.998

CmO2+1				
1	---	1	Cm(1)O(2)	278.999
CmO2+2				
2	---	1	Cm(1)O(2)	278.999
CmO2 (am. , s) Curium dioxide, amorphous				
0	---	2	Cm(1)O(2)	278.999
CmO2 (s) Curium dioxide				
0	---	2	Cm(1)O(2)	278.999
CO3-2 Carbonate ion				
-2	3812-32-6	1435	C(1)O(3)	60.0092
DTPA-5 Diethylenetrinitrilotetraacetate ion; 1,4,7-triazaheptane-1,1,7,7-pentaacetate ion; N,N-bis[2-[bis(carboxymethyl)amino]ethyl]glycinate ion; DTPA ion				
-5	---	343	C(14)H(18)N(3)O(10)	388.311
e-1 Electron				
-1	---	3399	E(1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C(10)H(12)N(2)O(8)	288.214
EDTMP-8 Ethylenediaminetetramethylphosphonate ion				
-8	---	134	C(6)H(12)N(2)O(12)P(4)	428.064
Etbis(ImMePhos)-4 Ethylenebis(iminomethylenephosphonate) ion				
-4	---	67	C(4)H(10)N(2)O(6)P(2)	244.081

F-1 Fluoride ion				
-1	16984-48-8	1090	F(1)	18.9984
F*3Benzoylacetone-1 Trifluorobenzoylacetone; Benzoyltrifluoroacetone				
-1	---	22	C(10)H(6)F(3)O(2)	215.152
Gly-1 Glycinate ion; Aminoacetate ion				
-1	---	1356	C(2)H(4)N(1)O(2)	74.0593
Glycolic-1 Glycolate ion				
-1	57122-18-6	342	C(2)H(3)O(3)	75.0440
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794
H+1_CDTA-4				
-3	---	39	C(14)H(19)N(2)O(8)	343.313
H+1_Citric-3				
-2	---	93	C(6)H(6)O(7)	190.109
H+1_DTPA-5				
-4	---	44	C(14)H(19)N(3)O(10)	389.319
H+1_EDTA-4				
-3	---	70	C(10)H(13)N(2)O(8)	289.222
H+1_EDTMP-8				
-7	---	11	C(6)H(13)N(2)O(12)P(4)	429.071
H+1_Etbis(ImMePhos)-4				
-3	---	20	C(4)H(11)N(2)O(6)P(2)	245.089

H+1_Gly-1 Glycine; Aminoacetic acid; Aminoethanoic acid; 2-Aminoethanoic acid				
0	---	82	C(2)H(5)N(1)O(2)	75.0672
H+1_HIMDA-2				
-1	---	14	C(6)H(10)N(1)O(5)	176.149
H+1_HOEDTA-3				
-2	---	28	C(10)H(16)N(2)O(7)	276.246
H+1_HydrazineNNDiAcet-2				
-1	---	19	C(4)H(7)N(2)O(4)	147.111
H+1_MePhos-2				
-1	---	6	C(1)H(4)O(3)P(1)	95.0150
H+1_N2COOEtIDA-3				
-2	---	10	C(7)H(9)N(1)O(6)	203.152
H+1_NTA-3				
-2	---	33	C(6)H(7)N(1)O(6)	189.125
H+1_OHMePhos-2				
-1	---	8	C(1)H(4)O(4)P(1)	111.014
H+1_PhosAcet-3				
-2	---	21	C(2)H(3)O(5)P(1)	138.017
H+1(2)_EDDPI-2 Ethylenebis(iminomethylenephosphinic acid); EDDPI; Ethylenediamine-N,N'-di(methylenephosphinic) acid				
0	---	5	C(4)H(14)N(2)O(4)P(2)	216.114
H+1(2)_EDTMP-8				
-6	---	13	C(6)H(14)N(2)O(12)P(4)	430.079
H+1(2)_Etbis(ImMePhos)-4				

-2	---	22	C(4)H(12)N(2)O(6)P(2)	246.097
H+1(2)_PhosAcet-3				
-1	---	5	C(2)H(4)O(5)P(1)	139.025
H+1(3)_Citric-3				
0	---	51	C(6)H(8)O(7)	192.125
H+1(3)_EDTMP-8				
-5	---	10	C(6)H(15)N(2)O(12)P(4)	431.087
H+1(3)_Etbis(ImMePhos)-4				
-1	---	5	C(4)H(13)N(2)O(6)P(2)	247.105
H+1(4)_EDTMP-8				
-4	---	7	C(6)H(16)N(2)O(12)P(4)	432.095
H+1(5)_EDTMP-8				
-3	---	5	C(6)H(17)N(2)O(12)P(4)	433.103
H2(g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H(2)	2.01588
H2O Water				
0	7732-18-5	5947	H(2)O(1)	18.0153
HIMDA-2				
-2	---	201	C(6)H(9)N(1)O(5)	175.141
HOEDTA-3 N-[2-[bis(carboxymethyl)amino]ethyl]-N-(2-hydroxyethyl)glycinate ion; N-(2-Hydroxyethyl)ethylenedinitrilotriacetate; HEDTA-3				
-3	14047-40-6	321	C(10)H(15)N(2)O(7)	275.238
HydrazineNNDiAcet-2 Hydrazine-N,N'-diacetate anion				
-2	---	80	C(4)H(6)N(2)O(4)	146.103

Lactic-1 Lactate ion				
-1	---	643	C(3)H(5)O(3)	89.0709
MeEDTA-4				
-4	---	57	C(11)H(14)N(2)O(8)	302.241
N2 (g) Nitrogen gas; Nitrogen				
0	7727-37-9	566	N(2)	28.0134
N2COOEtIDA-3				
-3	---	39	C(7)H(8)N(1)O(6)	202.144
N2COOPhenIDA-3				
-3	---	31	C(11)H(8)N(1)O(6)	250.188
N2PyridMeAsp-2 N-(2-Pyridylmethyl)-L-aspartate ion; N-(2-Pyridylmethyl)iminodiacetate ion				
-2	---	88	C(10)H(10)N(2)O(4)	222.200
N3-1 Azide ion				
-1	14343-69-2	176	N(3)	42.0201
Na (s) Sodium; Sodium, cubic				
0	7440-23-5	209	Na(1)	22.9900
NO2-1 Nitrite ion				
-1	14797-65-0	175	N(1)O(2)	46.0055
NO3-1 Nitrate ion				
-1	14797-55-8	573	N(1)O(3)	62.0049
NTA-3 Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion				

-3	28528-44-1	751	C(6)H(6)N(1)O(6)	188.117
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C(2)O(4)	88.0196
P3O9-3 Trimetaphosphate ion				
-3	---	22	O(9)P(3)	236.917
PO4-3 Phosphate ion; Phosphate ion				
-3	14265-44-2	1594	O(4)P(1)	94.9716
S(s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S(1)	32.0600
SCN-1 Thiocyanate ion; NCS-1 equivalent				
-1	302-04-5	785	C(1)N(1)S(1)	58.0777
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576
Squaric-2 Squarate ion				
-2	---	27	C(4)O(4)	112.042
Tartaric-2 Tartrate ion				

-2	---	393	$C(4)H(4)O(6)$	148.072
TriMDTA-4				
-4	---	107	$C(11)H(14)N(2)O(8)$	302.241